

ChE494

Entrepreneurship in Chemical Engineering

Midterm Exam

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Patent Search

It had been forty-five years since 1974 the first patent of portable bidet came up. Current patents include many types of portable bidet. The portable bidet that must be installed on flush toilet or bathtub like US10076215, US9194115, US7284285, D400328, US5377364 and many other similar patents. And portable bidet completely independent from flush toilet like the one in D545,425. This portable bidet in D545,425 has full independent water circulation system. But this feature makes it very difficult to carry due to its size. Even for a more advanced portal bidet described in US7543339 has dry system, difficulty on carrying still exists.

The more recent patent depicts portable bidet of small bottle and efficient sprayers like US7475438. And there are many portable bidets sold on Amazon are quite easy to carry but two obvious disadvantages of these bidet are squeeze bottle requires strength which makes it difficult for old person and fluid is controlled by hand which is not stable nor enough sometime. Only replaceable nozzle or sprayer is not enough to guarantee bidet's sanity condition.

Here is some information:

Must be installed on flush toilet	Must be installed on bathtub	Patents on hand-held Bidet or Related Apparatus
US10,076,215	US3,947,898	US10267026 (hemorrhoid treatment, large bottle required)
US9,194,115		
US7,543,339		US7,475,438(with sprayer of multiple direction)
US7,284,285		D545,425S (Independent Portable Bidet)
US6,192,527B1		10,213,799 (with multi-use portable hand-held sprayer)
US D400,328		
US5,734,997		US6751813 (Nozzle strayer protector)
US378,697		US7,155,754B2 (Device on generating jet of water)
US5,377,364		
US5,271,104		6,918,896B2: (user-powered portable bidet, big, difficult to carry)
US4,259,754		
US4,195,369		

This new patent has many advantages. User is freed from a specific type of bottle. Different kinds of bottle connectors described in the patent will make any water bottle compatible to this bidet including water bottle that could be purchased in the supermarket. This feature dramatically decreases the size of packing hence makes it easier to carry.

Compare to the existing hand-held bidet, the power of this bidet is guaranteed by using a controllable pump. The pump will promise enough fluid strength to meet different needs.

Replaceable supply tube guarantees the sanity of this bidet to prevent any possible infection.

Here is the patent search according to the features of this patent.

Medical tube:

Package:

For those flexible tubes, a proper package is required to main sterile before replacement. We could not afford any contamination of those tubes before usage because the purpose of having replaceable tubes is to protect supply tubes from any dirt. Patent US10,456,495B2 is a device of packaging for storing medical tubes (products). When package of reserved tubes in case is designed, this patent should be avoided.

Material:

As far as the material of tube, patent 10,376,671B2 is about producing a medical tube using a medical grade polyethylene with lubricity enhancing additive. US 10,299,757 B2 introduces a method of preparing medical tube with a special type of resin. US 10,266,693B2 contains a method of preparing medical tube with silicon elastomer composition. US10,264,786B2 claims an antiseptic composition which has broad spectrum antimicrobial and antifungal activity. This composition could be used on any medical tube. US 10,233,318 B2 describes a medical tube comprising polyolefin-based resin. US 10,183,098 B2 depicts a polytetrafluoroethylene tube for medical usage.

10,022,518B2 introduces a special type of formation of a kind of medical tube.

Being Foldable:

US 9,896,576 describes a medical tube made of polymer composition with high tolerance of bending and heating. Tolerance of bending will make this tube foldable.

US 7,955,676B2 describes medical tubes with balanced and proper properties including transparency, flexibility and elasticity. These will make this tube proper for portable

bidet.

US 7,582,079B2 introduces a medical tube high flexibility in bending. This tube could transmit torque to resist tension. This tube has small wall thickness.

US 6,846,535B2 depicts a kink resistant medical tube. Kink resistance is an important character for foldable tubes.

Water bottle connector:

One obvious advantage of this patent is different kinds of bottle connector. This feature makes user free from carrying a specific type of bottle. But in this patent US9,907,454B2 or US9,049,984, one kind of bottle adaptor which has similar shape with design of 21 Figure B is designed. Also in patent US 8,197,771 B2, a connector of similar mechanism as 24 Figure B was designed and patented already.

Potential Customer:

Old persons: Whose muscles and junctions start shrinking and can not properly use toilet tissue.

Baby: Whose private parts can not tolerate strong friction and need very careful cleaning.

Patient: Whose private parts needs anti-inflammation or antiseptic treatment. For example, hemorrhoid patient and pudendum infection patient.

Travelers: For travelers who often go to poor or undeveloped area where sanitary condition is bad, private parts requires extra cleaning.

Potential Strategic Partners:

Hospital or Clinic (Urology, Anorectal or Ophthalmology Department)

For patients with disease or infection on private parts. (I understand bidet is for private parts. But if we point the tube up, we could use it to rinse eyes.)

Nursing Home (Place for homeless old persons)

Old person can not use their arm properly so tools to help cleaning private parts will be necessary to promote their life quality.

Postpartum Confinement Center:

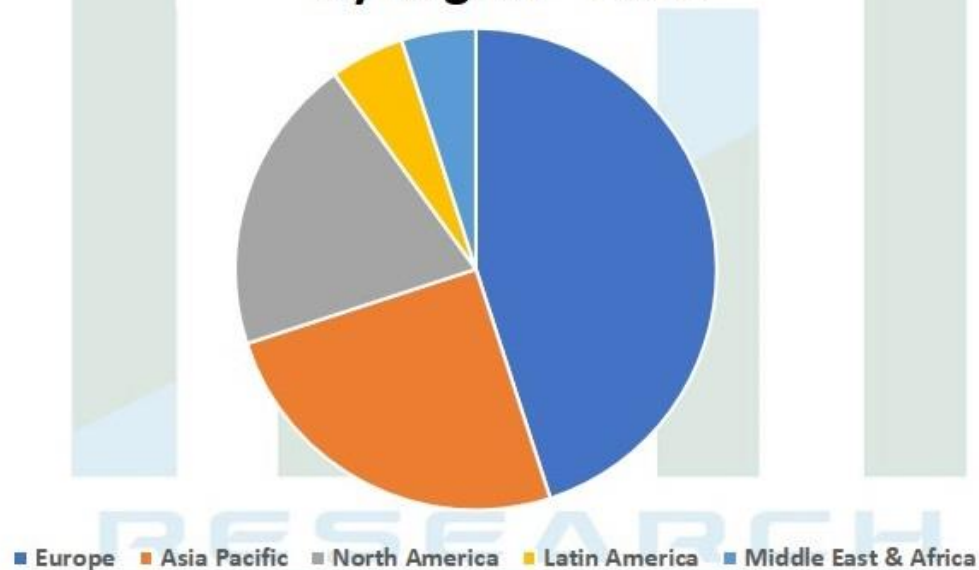
After giving birth to baby, women's vaginal and pudendum needs careful cleaning. Baby's private part could not tolerate strong friction.

UNHCR

Office of United Nation High Commission of Refugee is responsible for helping refugees and relieve the humanity crisis. Portable bidet will be helpful to solve sanity issue in undeveloped area and in refugee camp.

Portable Bidet Market Share

Portable Bidets For Travelers Market Share (%), By Region – 2019



Source: Research Nester

Competitive Pricing

Top 10 portable bidet brand and its price (all could be hand-held)

	Brand Name	Price
1	Brondell GoSpa Travel Bidet	9.99 USD
2	BioBidet The Palm TP-70 Portable Travel Bidet with 450ml Water Capacity	10.24USD
3	SHENG TIAN Travel Bidet Sprayer with 450ml Water Reservoir	9.99USD
4	Everore Portable Travel Bidet Sprayer	9.99USD
5	The Happy Brand Company Portable Bidet	14.95USD
6	Little Martin's Drawer Electric Portable Bidet	35.99USD
7	SINYO Portable Bidet with 450ml Capacity	9.2USD
8	Slopehill The Palm Handheld Portable Bidet	12.99 USD
9	TENZEPS Portable 600ml Bidet	11.98USD
10	QLav Personal Hygiene System	38.00 USD

Top 10 Bidet Seats

	Brand Name	Price
1	Smart Bidet Electrical Round	249.99 USD
2	TOTO S350e Elongated	942 USD
3	Astor Bidet Toilet Attachment	24.95 USD
4	Genie Bidet	58.99 USD
5	Luxe Neo 120 Attachment	34.95 USD
6	Superioe Bidet Attachment	45.94 USD
7	Luxe Bidet NEO 185	37.99 USD
8	Brondell Swash Elongated	279.00 USD
9	Luxe Bidet Neo 320	59.56 USD
10	Bio Bidet Ultimate Round	369.00 USD